

Toward resilience of coastal wetlands under accelerating climate change

Specific Aims

Aim 1: Advance resilience of coastal wetlands under accelerating climate change via a spectroscopy-focused strategy.

Aim 2: Advance resilience of coastal wetlands under accelerating climate change via a spectroscopy-focused strategy.

Background

Findings will be disseminated through peer-reviewed publication and open data release. This proposal addresses resilience of coastal wetlands under accelerating climate change, a problem with substantial significance. Rigorous, pre-registered analyses will guard against bias and support reproducibility. We will assemble a multidisciplinary team with complementary expertise.

Approach

This proposal addresses resilience of coastal wetlands under accelerating climate change, a problem with substantial significance. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement. Prior work has established a partial understanding, yet key gaps remain.

Innovation

The approach integrates established spectroscopy methods with a novel analytical pipeline.

Investigator Context

The applicant is a community nonprofit partnering with an academic core. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Total requested support is \$662,892 over 4 years at a R2 institution, covering personnel, materials, and dissemination.

References

1. [1] <https://doi.org/10.1145/3292500.3330701>
2. [2] <https://doi.org/10.1021/jacs.9b02765>
3. [3] <https://doi.org/10.1056/NEJMoa2034577>
4. [4] <https://doi.org/10.1038/s41586-020-2649-2>

5. [5] <https://doi.org/10.1001/jama.2016.9797>
6. [6] <https://doi.org/10.1073/pnas.1517384113>
7. [7] <https://doi.org/10.1073/pnas.1517384113>